AGRI 306 – DEVELOPMENT IN GRAINS AND CEREAL PRODUCTION AND MANAGEMENT

PEST AND DISEASE OF COFFEE IN THE PHILIPPINES

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DISEASES

Coffee Leaf Rust (CLR)

1. **Cause organism:** *Hemileia vastatrix*
2. **Symptoms:** Yellow-orange powdery spots on the underside of leaves; causes defoliation.
3. **Management:** Use resistant varieties (e.g., S795, Catimor); improve air circulation through pruning; apply copper-based fungicides.

H. vastatrix (orange spores) on a coffee leaf

Sooty Mold

1. **Cause organism :***Capnodium brasilense*.
2. **Symptoms:**A black, soot-like film covering leaves; usually a secondary result of scale insect or mealybug infestations.
3. **Management:** Control the sap-sucking insects (scales/mealybugs) first; the mold will then disappear. Application of Copper Fungicide

Sooty mold

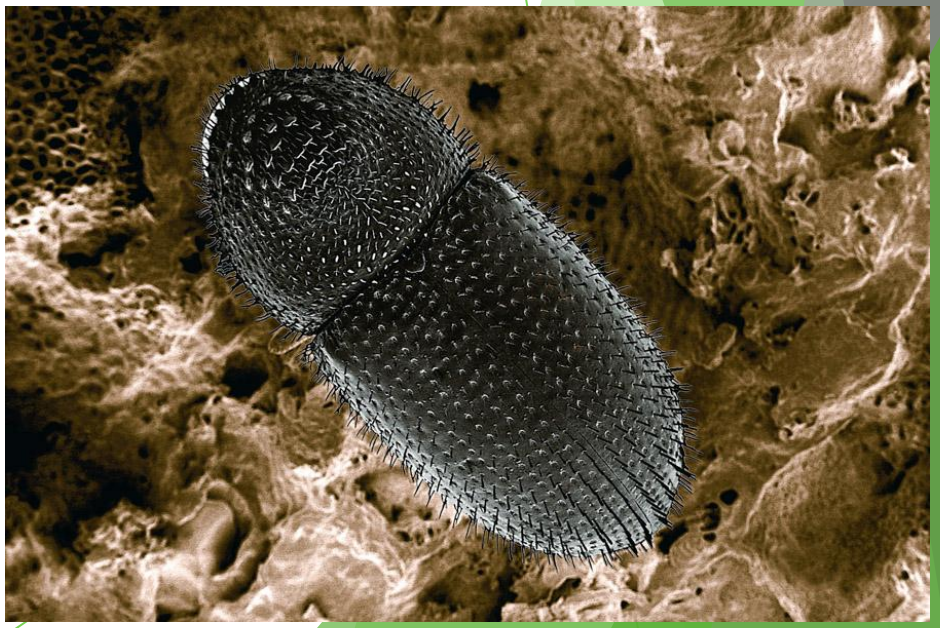
Pink disease

1. **Cause organism:** *Erythricium salmonicolor*
2. **Symptoms:**Pinkish fungal cobweb-like growth on stems and branches, leading to dieback.
3. **Management:** Prune and burn infected branches; ensure proper tree spacing for better airflow.

Pest

Coffee Berry Borer

1. **Scientific name:** *Hypothenemus hampei*
2. **Damage:**Small round hole at the tip of the coffee berry; Berries fall prematurely; Beans damaged inside, reducing quality; Powdery frass at entry hole
3. **Management:** Collect and destroy all fallen and infested berries; Timely and complete harvesting; Use traps to monitor infestation; Apply biological control (*Beauveria bassiana*); Chemical control only when infestation is high



Coffee Stem Borer

1. **Scientific name:** *Xylotrechus quadripes* (Chevrolat)
2. **Damage:** Wilting of branches; Small holes on the stem; Sawdust-like material at the base; Branch or whole plant death in severe cases.
3. **Management:** Regular inspection of stems; Remove and burn affected branches; Maintain proper shade management; Use recommended insecticides where necessary

Adult White stem borer

Green Coffee Scale

1. **Scientific name:** *Coccus viridis*
2. **Damage:** Small, green, oval-shaped insects that suck sap from young shoots and leaves. They secrete honeydew, which leads to **Sooty Mold**..
3. **Management:** Encouraging natural predators like ladybugs and lacewings; applying soap-based sprays or neem oil.

Scale insects

Thank you!!!